

① Selection Rules

(a) (i) $\langle n, 3, 1 | x | n, 1, 0 \rangle = 0$

This is zero because of the "dipole selection rule".
Dipole selection rule states that-

$$\langle \alpha_2 l_2 m_2 | z | \alpha_1 l_1 m_1 \rangle = 0 \text{ unless } \begin{matrix} l_2 = l_1 \pm 1 \\ m_2 = m_1 \end{matrix}$$

$$\text{and } \langle \alpha_2 l_2 m_2 | \hat{X} \text{ or } \hat{Y} | \alpha_1 l_1 m_1 \rangle = 0 \text{ unless } \begin{matrix} l_2 = l_1 \pm 1 \\ m_2 = m_1 \pm 1 \end{matrix}$$

(ii) In general

$$\langle \alpha_2 l_2 m_2 | T_k^q | \alpha_1 l_1 m_1 \rangle = 0$$

unless $l_2 = l_1 + k, l_1 + k - 1, \dots, |l_1 - k|$
 $m_2 = m_1 + q$.

$$\text{Now } xy = \sqrt{\frac{32\pi}{15}} \frac{r^2}{2i} [Y_2^2(\theta, \phi) - Y_2^{-2}(\theta, \phi)]$$

where $Y_l^m(\theta, \phi)$ are the spherical harmonics which
act as $T_{k=q}^{q=m}$.

Hence $\langle n, 2, 1 | xy | n, 1, 0 \rangle = \dots = \langle n, 2, 1 | T_2^2 - T_2^{-2} | n, 1, 0 \rangle$

Now the $l_2 = l_1 + k - 1$ is allowed.

But $m_2 = m_1 \pm 2$ is not satisfied.

$$(iii) \quad Z^2 = \sqrt{\frac{16\pi}{5}} \frac{r^2}{3} Y_2^0 + \sqrt{4\pi} \frac{r^2}{3} Y_0^0$$

(2)

$$\text{Thus } \langle \alpha_2 l_2 m_2 | Z^2 | \alpha_1 l_1 m_1 \rangle = 0$$

$$\text{unless } l_2 = l_1 + 2, \dots, |l_1 - 2|$$

$$m_2 = m_1$$

Thus

$\langle n10 | Z^2 | m00 \rangle$ can still be non-zero just based on these selection rules.

However,

$$\langle n10 | \hat{\Pi}^\dagger \Pi Z^2 \hat{\Pi}^\dagger \Pi | m00 \rangle = \langle n10 | Z^2 | m00 \rangle$$

$$\Pi Z^2 \hat{\Pi}^\dagger = Z^2$$

$$\hat{\Pi} | m00 \rangle = | m00 \rangle$$

$$\langle m10 | \hat{\Pi}^\dagger = - \langle m10 |$$

Parity on wavefunctions is $(-1)^l$.

$$\therefore \langle n10 | Z^2 | m00 \rangle = - \langle n10 | Z^2 | m00 \rangle$$

$$\therefore \langle n10 | Z^2 | m00 \rangle = 0.$$

Parity selection rule!

(b) (i) $\langle n | X^4 | n+3 \rangle = 0$

because of parity. X^4 is parity even.

$|n\rangle$ and $|n+3\rangle$ have opposite parity
(i.e... if one is even the other is odd).

(ii) $\langle n | X^5 | n+4 \rangle = 0$

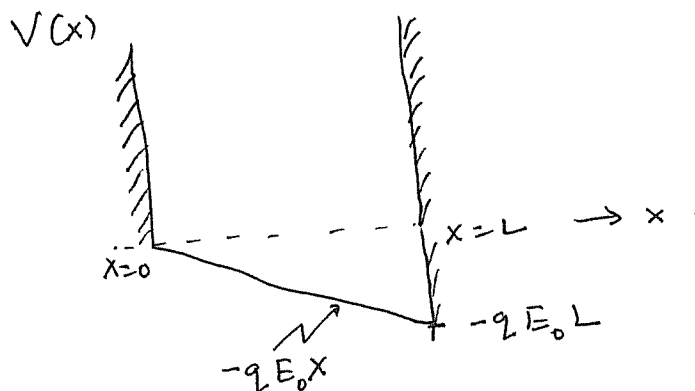
because of parity X^5 is parity odd
 $|n\rangle$ and $|n+4\rangle$ have same parity. Both are even or both are odd.

② The Hamiltonian of the system is

$$H = \frac{P^2}{2m} + V(x)$$

where

$$V(x) = \begin{cases} -qE_0 x & 0 \leq x \leq L \\ \infty & \text{otherwise} \end{cases}$$



We assume E_0 is very small, so that the perturbation (4)
is

$$V_{\text{pert}} = \begin{cases} -qE_0 x & 0 \leq x \leq L \\ 0 & \text{outside} \end{cases}$$

The unperturbed energy levels are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad n = 1, 2, 3, \dots$$

$$\psi_n(x) = \langle x | n \rangle = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

The energy shift is

$$E'_n = \int_0^L dx \frac{2}{L} \sin^2 \frac{n\pi x}{L} \cdot (-qE_0 x)$$

$$= -\frac{qE_0}{L} \int_0^L dx \left[1 - \cos \frac{2n\pi x}{L} \right] x$$

$$= -\frac{qE_0}{L} \int_{-L/2}^{L/2} dx \left[1 - \cos \left(\frac{2n\pi x}{L} + n\pi \right) \right] \left(x + \frac{L}{2} \right)$$

$$\boxed{E'_n = -\frac{qE_0 L}{2}}$$

All energy levels shift down by $-\frac{qE_0 L}{2}$.

③ We can write the Hamiltonian with the perturbation as ⑤

$$H = \frac{1}{2m} \begin{pmatrix} P_x & P_y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} P_x \\ P_y \end{pmatrix} + \frac{1}{2} m \omega^2 \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

(a) The ^{three} lowest lying states are $|n_x n_y\rangle$ $n_x=0$ $n_y=0$,
 $n_x=1$, $n_y=0$ and $n_x=0$ $n_y=1$.

$$E_{n_x n_y} = (n_x + n_y + 1) \hbar \omega.$$

$$\begin{aligned} |00\rangle &\rightarrow E_{00} = \hbar \omega \rightarrow \text{non-degenerate.} \\ |10\rangle &\rightarrow E_{10} = 2\hbar \omega \\ |01\rangle &\rightarrow E_{01} = 2\hbar \omega \end{aligned} \left. \vphantom{\begin{aligned} |10\rangle \\ |01\rangle \end{aligned}} \right\} \text{degenerate.}$$

(b) First order energy shift

$$E_{00}' = \langle 00 | \alpha m \omega^2 x y | 00 \rangle = 0.$$

E_{10}' E_{01}' can be calculated by diagonalizing

the matrix

$$\begin{pmatrix} \langle 10 | \alpha m \omega^2 x y | 10 \rangle & \langle 10 | \alpha m \omega^2 x y | 01 \rangle \\ \langle 01 | \alpha m \omega^2 x y | 10 \rangle & \langle 01 | \alpha m \omega^2 x y | 01 \rangle \end{pmatrix}$$

$\begin{matrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{matrix}$

$$\langle 01 | \alpha m \omega^2 x y | 10 \rangle = \frac{\hbar}{2m\omega} \cdot \alpha m \omega^2 = \frac{\alpha \hbar \omega}{2}$$

The zeroth order states are $\frac{1}{\sqrt{2}} \{ |10\rangle + |01\rangle \}$ and $\frac{1}{\sqrt{2}} \{ |10\rangle - |01\rangle \}$. The first order shift is $\pm \Delta$. (6)

where $\Delta = \frac{\alpha}{2} \hbar \omega$.

(c) In order to solve the problem exactly we define

$$\begin{pmatrix} x \\ y \end{pmatrix} = S \begin{pmatrix} x' \\ y' \end{pmatrix} \quad \text{where } S \text{ is a } 2 \times 2 \text{ orthogonal matrix to be determined shortly.}$$

$$\begin{pmatrix} p_x \\ p_y \end{pmatrix} = S \begin{pmatrix} p'_x \\ p'_y \end{pmatrix}$$

We rewrite the Hamiltonian in terms of x', y', p'_x, p'_y .

We get

$$H = \frac{1}{2m} (p'_x p'_y)^T \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} S \begin{pmatrix} p_x \\ p_y \end{pmatrix} + \frac{1}{2} m \omega^2 (x' y')^T \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix} S \begin{pmatrix} x' \\ y' \end{pmatrix}$$

we choose S such that

$$S^T \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix} S = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix} \quad S^T \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} S = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

or S diagonalises $M \equiv \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$ and S is orthogonal.

Since M is ^{real} symmetric this can be done.

α_1, α_2 are eigen values of M . i.e... the eigenvalue equation is

$$(1-\lambda)^2 - \alpha^2 = 0 \Rightarrow \lambda = 1 \pm \alpha.$$

$$\therefore \alpha_1 = 1 - \alpha \quad \text{and} \quad \alpha_2 = 1 + \alpha.$$

We don't even need to find S , since we know it can be calculated and with that S , P'_x, P'_y, x', y' become canonical variables. i.e...

$$[x', p'_x] = [y', p'_y] = i\hbar \quad [x', p'_y] = [y', p'_x] = 0.$$

The energy eigenvalues are

$$E_{n_x n_y} = \left(n_x + \frac{1}{2}\right)\hbar\omega\sqrt{1+\alpha} + \left(n_y + \frac{1}{2}\right)\hbar\omega\sqrt{1-\alpha}.$$

$$n_x=0 \quad n_y=0$$

$$E_{00} = \frac{1}{2}\hbar\omega [\sqrt{1+\alpha} + \sqrt{1-\alpha}]$$

$$n_x=1 \quad n_y=0$$

$$E_{10} = \hbar\omega\sqrt{1+\alpha} + \frac{1}{2}\hbar\omega [\sqrt{1+\alpha} + \sqrt{1-\alpha}]$$

$$n_x=0 \quad n_y=1$$

$$E_{01} = \hbar\omega\sqrt{1-\alpha} + \frac{1}{2}\hbar\omega [\sqrt{1+\alpha} + \sqrt{1-\alpha}].$$

up to first order in α we see

$$E_{00} = \hbar\omega$$

$$E_{10} = 2\hbar\omega + \frac{\alpha}{2}\hbar\omega.$$

$$E_{01} = 2\hbar\omega - \frac{\alpha}{2}\hbar\omega$$

as we found earlier.

(8)

17.2.6

$$\psi_{2,1,0} = \left(\frac{1}{32\pi a_0^3} \right)^{1/2} \frac{r}{a_0} e^{-r/2a_0} \cos\theta$$

$$\psi_{1,0,0} = \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-r/a_0}$$

$$\langle 210 | Z | 100 \rangle = \int d^3r \left(\frac{1}{32\pi a_0^3} \right)^{1/2} \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-3r/2a_0} \frac{r^2 \cos^2\theta}{a_0}$$

$$= \frac{2\pi}{\pi a_0^4 \sqrt{32}} \int_0^\infty dr r^4 e^{-3r/2a_0} \int_0^\pi d\theta \sin\theta \cos^2\theta$$

$$= \frac{1}{2a_0^4} \frac{1}{\sqrt{2}} \left(\frac{2a_0}{3} \right)^5 \int_0^\infty dt t^4 e^{-t} \left(\frac{-\cos^3\theta}{3} \right)_0^\pi$$

$$= \frac{2^4}{3^5 \sqrt{2}} a_0 \cdot 4! \cdot \frac{2}{3} = \frac{2^8}{3^5 \sqrt{2}} a_0$$

$$|\langle 210 | Z | 100 \rangle|^2 = \frac{2^{15}}{3^{10}} a_0^2$$

17.2.7

$$H' = -q f X$$

We want to find $\hat{\Omega}$ such that $[\Omega, H^0] = H'$

It is easily to check

$$\hat{\Omega} = \frac{-iqf}{m\hbar\omega^2} \hat{P}$$

Then

$$\begin{aligned} E_n^2 &= \langle n | H, \hat{\Omega} | n \rangle - \langle n | H' | n \rangle \langle n | \hat{\Omega} | n \rangle \\ &= \frac{iq^2 f^2}{m\hbar\omega^2} \langle n | \hat{X} \hat{P} | n \rangle \end{aligned}$$

$$\hat{X} = \left(\frac{\hbar}{2m\omega} \right)^{1/2} (a + a^\dagger)$$

$$\hat{P} = \left(\frac{\hbar m \omega}{2} \right)^{1/2} \frac{1}{i} (a - a^\dagger)$$

9

We can see that

$$E_n^2 = - \frac{q^2 f^2}{2m\omega^2}$$

which matches (17.2.6)
as expected!